

Improper integrals – 3

Improper integrals depending on a parameter

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Theoretical minimum.

Definition. $\int_a^\infty f(x, t) dx$ converges *uniformly* on a set T if $\sup_{t \in T} \left| \int_b^\infty f(x, t) dx \right| \rightarrow 0$ as $b \rightarrow \infty$. Pointwise convergence for every t says nothing about the behaviour of the integral as a function of t ; uniform convergence says everything.

Theorem 1 (criteria). Cauchy: uniform convergence is equivalent to $\sup_t \left| \int_{b_1}^{b_2} f(x, t) dx \right| \rightarrow 0$. Weierstrass: if $|f(x, t)| \leq g(x)$ with $\int_a^\infty g < \infty$, the convergence is uniform. Dirichlet: if $\int_a^b f(x, t) dx$ is bounded uniformly in b and t , and $g(x, t)$ is monotone in x and tends to 0 uniformly in t , then $\int_a^\infty fg$ converges uniformly. Abel: the same with g monotone and uniformly bounded and $\int_a^\infty f(x, t) dx$ uniformly convergent.

Theorem 2 (what uniform convergence buys). Let $F(t) = \int_a^\infty f(x, t) dx$ converge uniformly on $[c, d]$ with f continuous. Then F is continuous, and $\int_c^d F = \int_a^\infty \left(\int_c^d f dt \right) dx$. If moreover $\partial_t f$ is continuous and $\int_a^\infty \partial_t f dx$ converges uniformly, then F is differentiable and $F'(t) = \int_a^\infty \partial_t f(x, t) dx$.

Fact 1 (the standard counterexample). $\int_0^\infty te^{-tx} dx = 1$ for every $t > 0$ and $= 0$ for $t = 0$: the convergence is not uniform near $t = 0$, and the limit of the integrals is not the integral of the limit. Every failure of the Feynman trick looks like this example.

Fact 2 (what this sheet repairs). In Integrals – 3 the parameter trick was applied to integrals over $[0, \infty)$ without justification, and in Definite integrals – 8 only the proper case on a rectangle was proved. The theorems above are exactly the missing link, and the first two problems supply it for the two computations used most often.

1. Prove the Cauchy criterion and the Weierstrass test, and decide the uniform convergence of $\int_0^\infty e^{-tx} \frac{\sin x}{x} dx$ on $[0, \infty)$, of $\int_0^\infty t e^{-tx} dx$ on $[0, 1]$ and of $\int_0^\infty \frac{\sin(tx)}{x} dx$ on $[\delta, \infty)$ and on $\mathbb{R}$.
2. Supply the justification that was taken on trust in Integrals – 3: for $F(t) = \int_0^\infty e^{-tx} \frac{\sin x}{x} dx$ prove that F is continuous on $[0, \infty)$ and differentiable on $(0, \infty)$ with $F'(t) = -\frac{1}{1+t^2}$, and deduce $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$. Which of the two tests of Theorem 1 is used at $t = 0$?
3. Prove that $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$ converges for $s > 0$, that the convergence is uniform on every $[a, b] \subset (0, \infty)$ and is not uniform on $(0, 1]$, and that Γ is infinitely differentiable with $\Gamma^{(k)}(s) = \int_0^\infty x^{s-1} \ln^k x e^{-x} dx$.
4. Give an example of a family with $\int_0^\infty f(x, t) dx$ and $\int_0^\infty \partial_t f(x, t) dx$ both convergent for every t , for which nevertheless $F'(t) \neq \int_0^\infty \partial_t f dx$; and an example where F is discontinuous although f is continuous and the integral converges for every t .
5. (SEEMOUS) Let $n \geq 1$ be an integer and let $I_n = \int_0^\infty \frac{\arctan x}{(1+x^2)^n} dx$. Prove that
 - a) $\sum_{n=1}^\infty \frac{I_n}{n} = \frac{\pi^2}{6}$;
 - b) $\int_0^\infty \arctan x \cdot \ln\left(1 + \frac{1}{x^2}\right) dx = \frac{\pi^2}{6}$.